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Batch-51

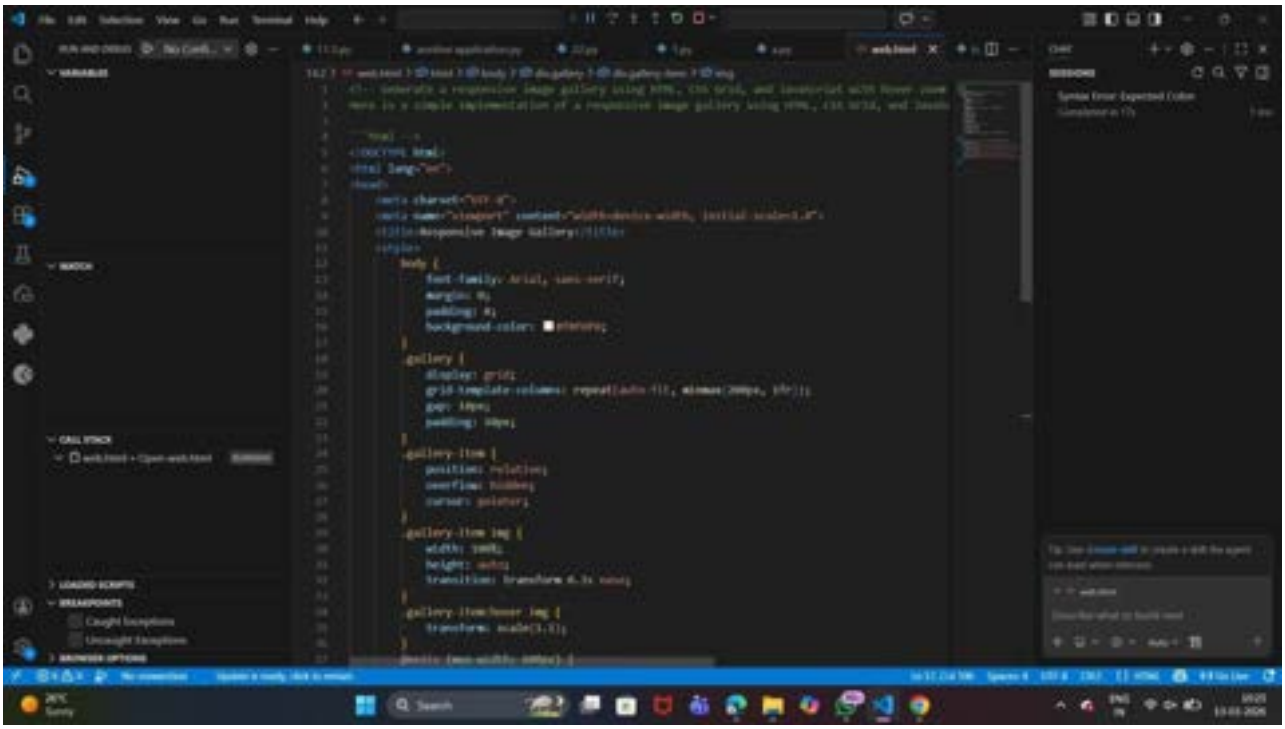
ASSIGNMENT 14.5

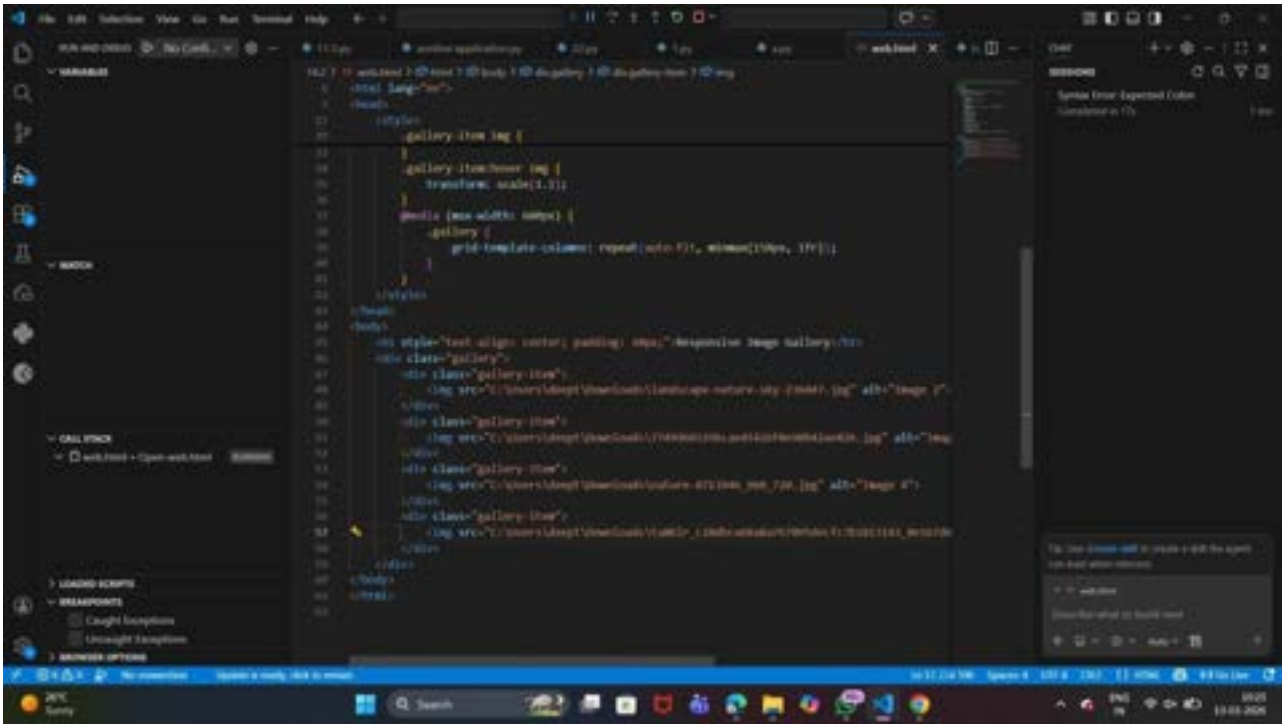
Web Design Application – AI-Assisted HTML/CSS/JS Generation

Task Description #1 (Image Gallery Website):

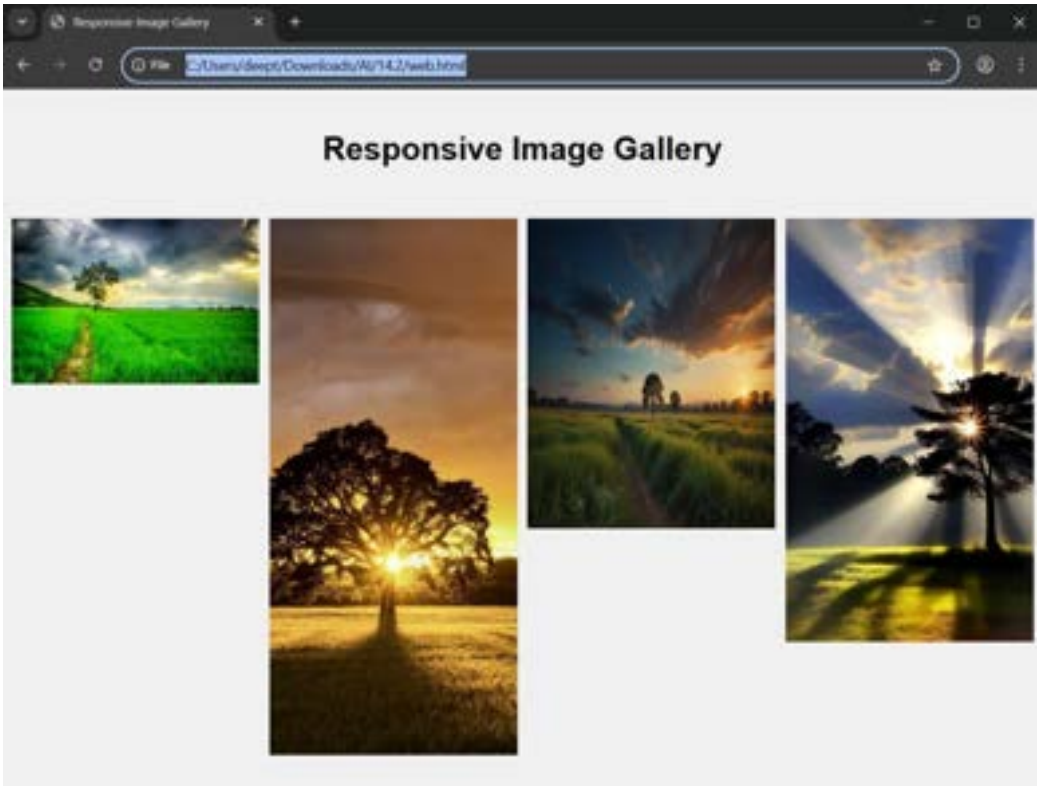
PROMPT: Generate a responsive image gallery using HTML, CSS Grid, and JavaScript with hover zoom effects and mobile-friendly layout.

CODE:





OUTPUT:



Explanation

• CSS Grid arranges images in a structured grid.
• Hover zoom effect improves visual interaction.
• Responsive design adjusts layout for different devices.

Task Description #2 (Profile Card Generator):

PROMPT: Create a modern profile card using HTML, CSS Flexbox, and JavaScript showing user image, name, role, and social media links with hover animation.

CODE:

OUTPUT:

Explanation

• Flexbox aligns card elements properly.

• Modern colors and fonts improve UI design.

• Hover animation adds interactive effects.

Task Description #3 (Develop a dynamic shopping cart system):

Prompt : Develop a dynamic shopping cart using HTML, CSS, and JavaScript that allows adding, removing, and updating items with real-time total price and localStorage support.

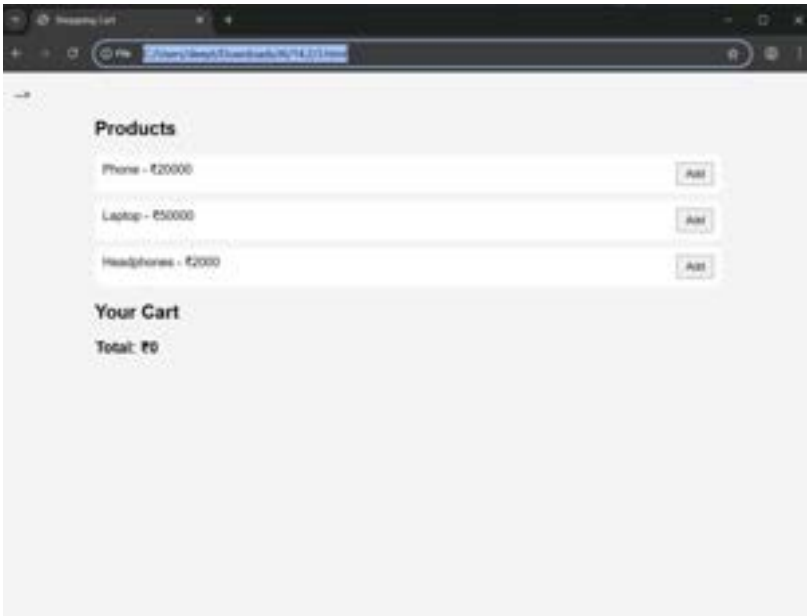
Code :

```html
<!-- Develop a dynamic shopping cart using HTML, CSS, and JavaScript that allows adding, remo
Manage cart state using JavaScript objects.
o Persist cart data in localstorage.
o Update total price in real-time as items are added, removed, or updated.
Here is a simple implementation of a dynamic shopping cart using HTML, CSS, and JavaScript the
  html - a -->
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Shopping Cart</title>

<style>
body{
    font-family: Arial;
    background:#f4f4f4;
    margin:20px;
}

.container{
    max-width:800px;
    margin:auto;
}

.product, .cart-item{
    display:flex;
    justify-content:space-between;
    background:white;
    padding:10px;
    margin:10px 0;
    border-radius:6px;
}

button{
    padding:5px 10px;
    cursor:pointer;
```

```html
<html lang="en">
<head>
<style>
}
</style>
</head>

<body>

<div class="container">

<h2>Products</h2>

<div class="product">
<span>Phone - ₹20000</span>
<button onclick="addToCart('Phone',20000)">Add</button>
</div>

<div class="product">
<span>Laptop - ₹50000</span>
<button onclick="addToCart('Laptop',50000)">Add</button>
</div>

<div class="product">
<span>Headphones - ₹2000</span>
<button onclick="addToCart('Headphones',2000)">Add</button>
</div>

<h2>Your Cart</h2>
<div id="cart"></div>

<div class="total">
Total: ₹<span id="total">0</span>
</div>

</div>
```

```
<html lang="en">
<body>
<script>
function updateCart(){
let total = 0;

for(let item in cart){

let price = cart[item].price;
let qty = cart[item].qty;

total += price * qty;

cartDiv.innerHTML +=
`<div class="cart-item">
<span>${item} - ₹${price}</span>

<input type="number" value="${qty}" min="1"
onchange="changeQty('${item}',this.value)">

<button onclick="removeItem('${item}')">Remove</button>
</div>
`;
}

document.getElementById("total").innerText = total;

localStorage.setItem("cart", JSON.stringify(cart));
}

updateCart();

</script>

</body>
</html>
```

Output :



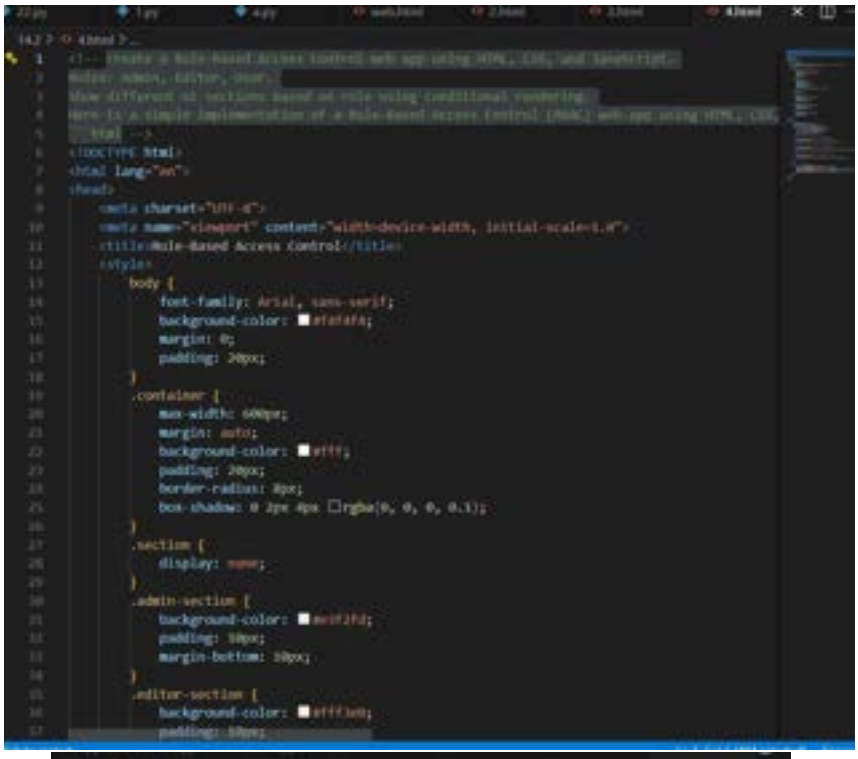
Explanation

- JavaScript objects manage cart items.
- Total price updates automatically.
- localStorage saves cart data after refresh.

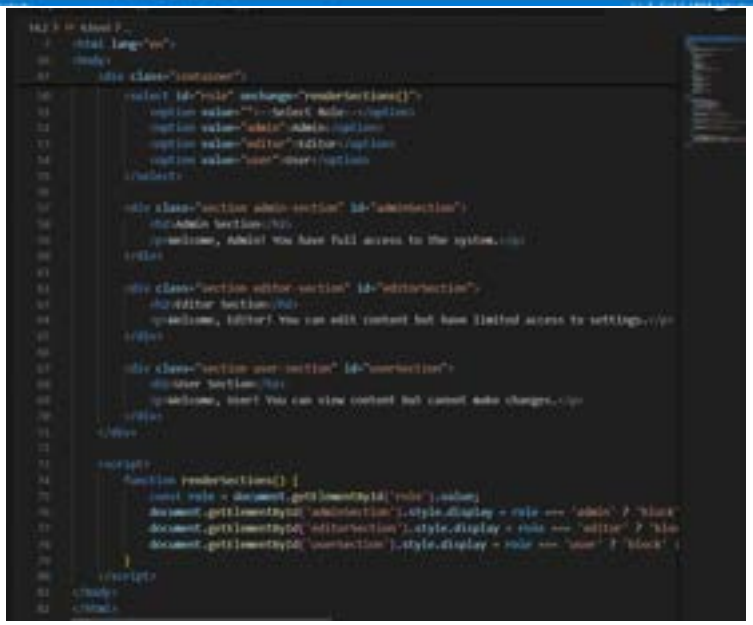
Task description #4 (Role-Based Access Control Application):

Prompt : Create a role-based web application using HTML, CSS, and JavaScript with Admin, Editor, and User roles and conditional UI rendering and Design access control logic,Maintain secure state transitions,Create scalable UI components.

Code :

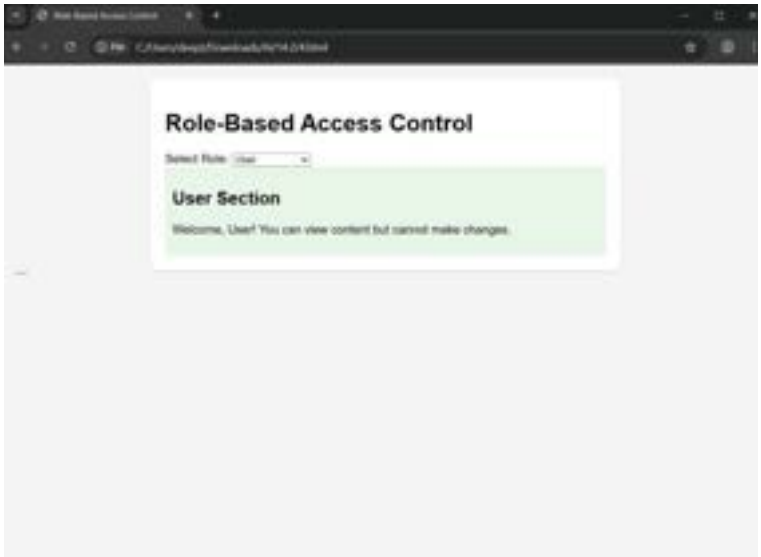


```
<!-- Create a Role-Based Access Control web app using HTML, CSS, and JavaScript.
Roles: Admin, Editor, User.
Show different UI sections based on role using conditional rendering.
Here is a simple implementation of a Role-Based Access Control (RBAC) web app using HTML, CSS,
  and JS. -->
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Role-Based Access Control</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            background-color: #f4f4f4;
            margin: 0;
            padding: 20px;
        }
        .container {
            max-width: 600px;
            margin: auto;
            background-color: #fff;
            padding: 20px;
            border-radius: 8px;
            box-shadow: 0 2px 4px rgba(0, 0, 0, 0.1);
        }
        .section {
            display: none;
        }
        .admin-section {
            background-color: #e1f2fd;
            padding: 10px;
            margin-bottom: 10px;
        }
        .editor-section {
            background-color: #fff3e0;
            padding: 10px;
```



```
<html lang="en">
<body>
    <div class="container">
        <select id="role" onchange="renderSections()">
            <option value="">-- Select Role --</option>
            <option value="admin">Admin</option>
            <option value="editor">Editor</option>
            <option value="user">User</option>
        </select>

        <div class="section admin-section" id="adminSection">
            <h2>Admin Section</h2>
            <p>Welcome, Admin! You have full access to the system.</p>
        </div>

        <div class="section editor-section" id="editorSection">
            <h2>Editor Section</h2>
            <p>Welcome, Editor! You can edit content but have limited access to settings.</p>
        </div>

        <div class="section user-section" id="userSection">
            <h2>User Section</h2>
            <p>Welcome, User! You can view content but cannot make changes.</p>
        </div>
    </div>

    <script>
        function renderSections() {
            const role = document.getElementById('role').value;
            document.getElementById('adminSection').style.display = role === 'admin' ? 'block'
            document.getElementById('editorSection').style.display = role === 'editor' ? 'blo
            document.getElementById('userSection').style.display = role === 'user' ? 'block' :
        }
    </script>
</body>
</html>
```

Output :



Explanation

- Different roles have different permissions.
- UI changes based on user role.
- Improves security and access control.

Task Description #5 (Multi-Language Website-Design a multilingual website):

Prompt : Design a multilingual website using HTML, CSS, and JavaScript with dynamic language switching and language preference stored in localStorage and Structure translation files,Handle RTL/LTR layouts,Improve accessibility.

Code :

```html
<!-- Develop a multilingual website using HTML, CSS, and JavaScript.
Allow dynamic language switching.
Store translations in JSON and save language preference in localStorage.
Here is a simple implementation of a multilingual website using HTML, CSS, and JavaScript that
```html ... >
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Multilingual Website</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            background-color: #f4f4f4;
            margin: 0;
            padding: 20px;
        }
        .container {
            max-width: 600px;
            margin: auto;
            background-color: #fff;
            padding: 20px;
            border-radius: 8px;
            box-shadow: 0 2px 4px rgba(0, 0, 0, 0.1);
        }
        .language-selector {
            margin-bottom: 20px;
        }
    </style>
</head>
<body>
    <div class="container">
        <h1 id="title">Welcome to our website!</h1>
        <p id="description">This is a simple multilingual website example.</p>
        <div class="language-selector">
            <label for="language">Select Language:</label>
```

```html
<html lang="en">
<body>
    <div class="container">
    </div>

    <script>
        const translations = {
            en: {
                title: "Welcome to our website!",
                description: "This is a simple multilingual website example."
            },
            es: {
                title: "¡Bienvenido a nuestro sitio web!",
                description: "Este es un ejemplo de sitio web multilingüe simple."
            },
            fr: {
                title: "Bienvenue sur notre site web!",
                description: "Ceci est un exemple de site web multilingue simple."
            }
        };

        function switchLanguage() {
            const language = document.getElementById('language').value;
            document.getElementById('title').textContent = translations[language].title;
            document.getElementById('description').textContent = translations[language].descri
            localStorage.setItem('preferredLanguage', language);
        }

        // Load preferred language on page load
        window.onload = function() {
            const preferredLanguage = localStorage.getItem('preferredLanguage') || 'en';
            document.getElementById('language').value = preferredLanguage;
            switchLanguage();
        };
    </script>
</body>
</html>
```

Output :
```

Explanation:

- JSON files store translation data.
- Users can switch languages dynamically.
- Language preference is saved for future visits.

Task Description #6 (Online Examination System-Create an online exam interface.):

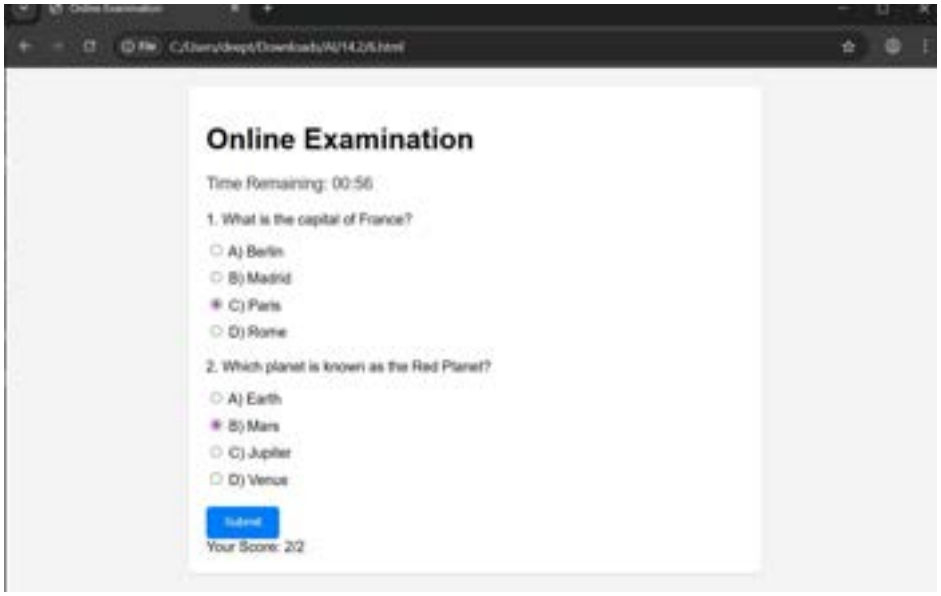
Prompt : Create an online examination interface using HTML, CSS, and JavaScript with MCQ questions, countdown timer, and single submission system and Manage exam state,Improve exam UX,Ensure accessibility.

Code :

```html
<html lang="en">
<head>
    </style>
</head>
<body>
    <div class="container">
        <h1>Online Examination</h1>
        <div class="timer" id="timer">Time Remaining: 01:00</div>
        <div class="question">
            <p>1. What is the capital of France?</p>
            <ul class="options">
                <li><input type="radio" name="q1" value="A"> A. Berlin</li>
                <li><input type="radio" name="q1" value="B"> B. Madrid</li>
                <li><input type="radio" name="q1" value="C"> C. Paris</li>
                <li><input type="radio" name="q1" value="D"> D. Rome</li>
            </ul>
        </div>
        <div class="question">
            <p>2. Which planet is known as the Red Planet?</p>
            <ul class="options">
                <li><input type="radio" name="q2" value="A"> A. Earth</li>
                <li><input type="radio" name="q2" value="B"> B. Mars</li>
                <li><input type="radio" name="q2" value="C"> C. Jupiter</li>
                <li><input type="radio" name="q2" value="D"> D. Venus</li>
            </ul>
        </div>
        <button class="submit-btn" id="submitBtn">Submit</button>
        <div id="result"></div>
    </div>
    <script>
        let timeLeft = 60;
        const timerElement = document.getElementById('timer');
        const submitBtn = document.getElementById('submitBtn');
        const resultElement = document.getElementById('result');
        let timerInterval;

        function startTimer() {
```

```html
<html lang="en">
<body>
    <script>
        function startTimer() {
            timerInterval = setInterval(() => {
                    submitExam();
                } else {
                    timeLeft--;
                    const minutes = Math.floor(timeLeft / 60).toString().padStart(2, '0');
                    const seconds = (timeLeft % 60).toString().padStart(2, '0');
                    timerElement.textContent = `Time Remaining: ${minutes}:${seconds}`;
                }
            }, 1000);
        }

        function submitExam() {
            submitBtn.disabled = true;
            const q1Answer = document.querySelector('input[name="q1"]:checked')?.value;
            const q2Answer = document.querySelector('input[name="q2"]:checked')?.value;
            let score = 0;

            if (q1Answer === 'C') score++;
            if (q2Answer === 'B') score++;

            resultElement.textContent = `Your Score: ${score}/2`;
        }

        submitBtn.addEventListener('click', () => {
            clearInterval(timerInterval);
            submitExam();
        });

        startTimer();
    </script>
</body>
</html>
```

Output :



Explanation :

- Timer controls exam duration.
- JavaScript manages exam state.
- Prevents multiple submissions.